Al-Balqa Applied University

جامعة البلقاء التطبيقية

Midterm Exam

23/30

Course Title	General Chemistry lab		Course Code	30206111	
Semester	2nd		Academic Year	2025-2026	
Exam Date	27/4/2026	Day	Monday	Time	2:30-3:20
Student Name	[redacted]		udent ID No.	Section	

اسم المدرس :

± GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:**

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1 – 10	Multiple Choice	CLIO 1-5	33.3 %	10		8
11 - 15	Assay	CLIO 2-5	66.7 %	20		15
Total			100 %	30		
Moderation Review						
Comments						
Marks Reviewed by					Signature & Date	

Question	1	2	3	4	5
Answer	B	B	D	D	B

PArT1 (10 marks)

Choose the answer of the following questions and transfer it to the table in CAPITAL letters

Q1: Which piece of safety equipment should be used to protect your eyes during a chemical experiment?

A. Lab coat
B. Safety goggles
C. Fire extinguisher
D. First-aid kit

Q2: What is the best practice when taking chemicals from a stock container?

A. Take a large amount to avoid repeated trips
B. Take exactly all the chemical needed at once
C. Take very small amounts, especially at the beginning
D. Take chemicals only after the experiment starts

Q3: Which of the following compounds is generally insoluble in water?

A. $NaCH_3COO$
B. NH_4Cl
C. K_2SO_4
D. PbI_2

Q4: Which of the following factors would result in a higher boiling point for a substance?

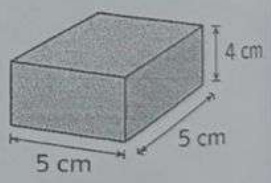
A. Weaker intermolecular forces
B. Lower external atmospheric pressure
C. Lower molar mass
D. Strong hydrogen bonding between molecules

Q5: What is the main purpose of placing the hot crucible in a desiccator before weighing?

A. To speed up the cooling process
B. To prevent absorption of moisture from the air
C. To allow more water to evaporate
D. To change the color of the salt

PART2 (20 MARKS) : Answer the following questions :

Q6- A rectangular solid metal block has a **length of 5 cm**, a **width of 5 cm**, and a **height of 4 cm**. The **mass of the block is 640 g**.*(You should show all steps and calculations.)(5marks)*



A) Calculate the **volume** of the solid.(2marks)

V = length × width × height

5cm × 5cm × 4cm = $V = 100\ cm^3$ (2)

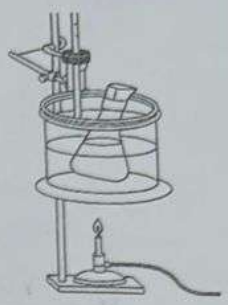
B) Calculate the **density** of the solid.(2 marks)

$d = \frac{m}{V} = \frac{64.0\,g}{100\,cm^3}$

$d = 6.4\ g/cm^3$ (2)

C) State the **correct unit** of density.(1 marks)

g/cm^3 or g/mL (1)



Q7: for the given setup, answer the following: (5marks)

Awt g/mol: C= 12 ,H = 1 , Cl= 35.5 / R=0.08206L.atm/K.mol

A) What is the name of the experiment represented by this setup?(2 marks)

boiling point temperature B.pt

B) A sample of an unknown liquid is completely vaporized in a container with the following data find the molar mass of this liquid.(2 marks)

Volume	225 mL
Temperature	92 °C
Pressure	770 torr
Mass of vapor	0.860 g

$M.m = \frac{mRT}{PV}$

$V = 0.225\ L$

$T = 92 + 273 = 365\ K$

$P = 1.013\ atm$

$= \frac{0.86 \times 0.0826 \times 365}{1.013 \times 0.225}$

$M.w = 113$ (exactly = 112.99) (2)

1 atm — 760

x — 770

$x = \frac{770}{760}$

C) Based on your calculations determine the unknown liquid (C_6H_5Cl or C_7H_8) ?(1mark)

$6 \times 12 + 5 + 35.5 = 112.5$ (1)

C_6H_5Cl (1)

8

Q8: A sample of a hydrated metal sulfate, $MSO_4 \cdot xH_2O$, has a mass of 5.062 g. After heating to constant mass, the remaining anhydrous salt weighs 2.408 g. (5 marks)

A) Calculate the mass of water lost during heating. (1 mark)

$5.062 - 2.408 = 2.654$ (1)

B) Determine the mass percent of water in the hydrate. (2 marks)

$\%\ H_2O = \frac{2.654}{5.062} \times 100 = 52.42\ \%$ (2)

C) Find the value of x and write the empirical formula of the hydrate. (Molar mass of MSO_4 = 120.37 g/mol) (2 marks)

$X = \frac{\text{moles of } H_2O}{\text{moles of anhydrites}} = \frac{2.654/18}{2.408/120.37} = 7.3 \approx 7$ (2)

$MSO_4 \cdot 7H_2O$

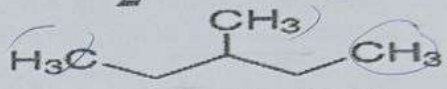
Q9: Arrange the following compounds in **increasing order of boiling point** (lowest → highest). (5 Marks)

1 (7)



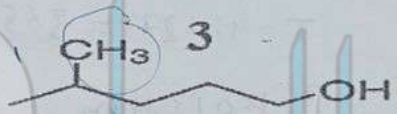
n-pentane

2 (3) (5)



2-methylbutane

3



Diethel ether

4



1-butanol (5)

5



Pentan-2-one (4)

Lowest B.p				Highest B.p
5	2	4	3	1

52431

(5)